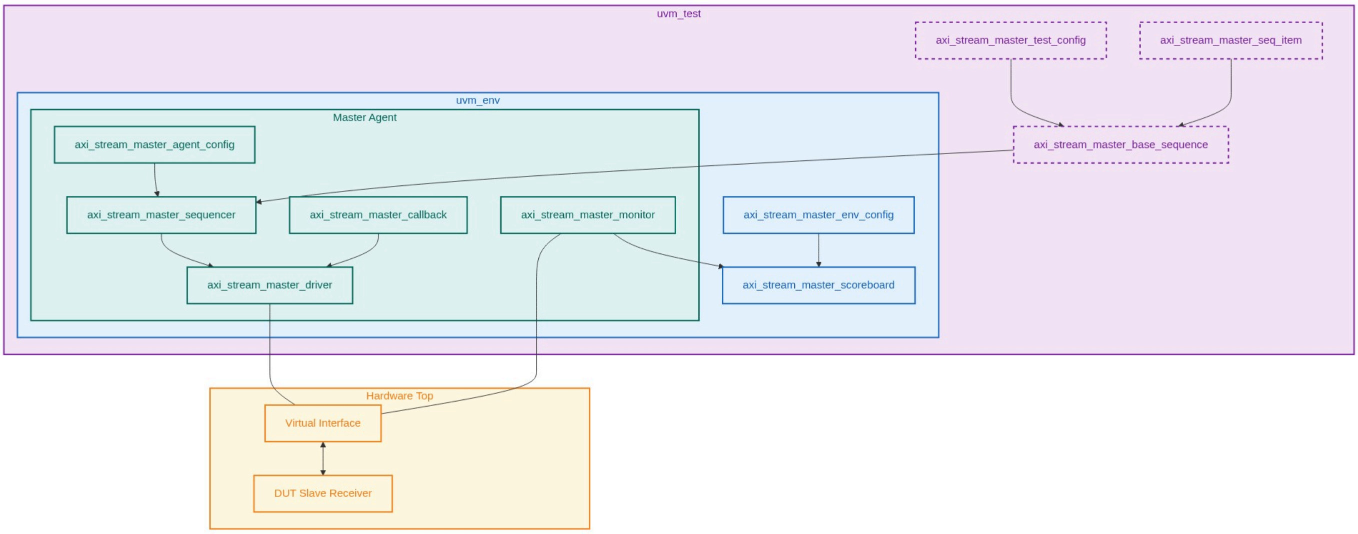
Verification Plan Report

Project: AXI-Stream Master Verification IP

Version: 20.0

Verification Role: Master

1. UVM Environment Architecture



2. Test Requirements

Index	Feature/Sub	Name	Description
REQ_MST_01	Handshake (TVALID Stability)	Continuous TVALID Post-Assertion	Hardware Logic: The valid data bytes and control information from the Transmitter must remain unchanged once TVALID has been asserted. The Transmitter is not permitted to wait until TREADY is asserted before asserting TVALID. Premature dropping of TVALID or changing data creates race conditions at the receiver boundaries. Functional constraint guarantees TVALID remains HIGH and TDATA/TSTRB/TKEEP/TID/TDEST/TLAST are completely locked until TREADY is sampled HIGH. Negative scenarios will explicitly attempt to force invalid transitions to test VIP monitor error detection.

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REQ_MST_02	Handshake (TREADY Delay)	Receiver Delay Handling	Hardware Logic: A Receiver is permitted to wait for TVALID to be asserted before asserting TREADY. The DUT Slave might insert arbitrary wait states, or single-cycle completions. Generate variable delays from the Master VIP. Since the VIP is Master, it doesn't control TREADY directly, but the agent's reactive sequence must verify that the DUT accepts back-to-back zero-delay bursts and max-wait-state bursts without packet loss.
REQ_MST_03	Data Routing (Stream Interleaving)	TID/TDEST Integrity	Hardware Logic: Transfers that have the same TID and TDEST values are from the same stream. Interleaving of transfers in different streams is permitted on a per transfer basis and is not limited to TLAST boundaries. VIP will generate heavily interleaved traffic across multiple TID streams. Constraint verification to guarantee no mixed packets unless IDs differ. Performance profiling of structural throughput under 100% interleaving.
REQ_MST_04	Data Packing (Null & Position Bytes)	TKEEP and TSTRB Rules	Hardware Logic: TKEEP HIGH indicates payload transport. TSTRB HIGH indicates a data byte, LOW indicates a position byte. A null byte has TKEEP LOW. The DUT must correctly extract data and drop null bytes. Generate completely randomized TKEEP and TSTRB arrays, creating continuous aligned, continuous unaligned, and sparse streams. Negative constraint: TSTRB HIGH while TKEEP LOW is an explicit violation and must trigger an error.
REQ_MST_05	Data Packing (Packet Boundaries)	TLAST Framing	Hardware Logic: TLAST indicates the boundary of a packet. A packet can be 1 beat or deeply fragmented. A transfer can have TLAST asserted but contain zero data bytes to push buffers. Verify the DUT handles 1-beat packets, max-legnth packets, and null-termination packets (TLAST=1, all TKEEP=0). Check DUT sideband routing to ensure entire frame alignment.
REQ_MST_06	Reset (Mid-Transfer Reset)	Asynchronous Reset Recovery	Hardware Logic: ARESETn is active-LOW. During reset, TVALID must be driven LOW. All other signals can be driven to any value. Reset can happen asynchronously. Sequence will inject ARESETn midway through an interleaved packet burst. Verify all VIP components flush sequence queues, drop pending transactions, and resume gracefully post-reset without protocol deadlocks.

REQ_MST_07	AXI5 (Wake-up Signaling)	TWAKEUP Assertion Latency	Hardware Logic: TWAKEUP indicates activity. It must be driven before or with TVALID. If driven on the same cycle, TWAKEUP must remain asserted until TREADY. Constrain Master sequence to assert TWAKEUP randomly between 1 and 20 cycles before TVALID. Assert both on the exact same cycle. Drop TWAKEUP synchronously after packet completion.
REQ_MST_08	AXI5 (Parity Protection)	Odd Parity Verification	Hardware Logic: Error detection operates using odd parity signals (TVALIDCHK, TDATACHK, etc). Each parity bit covers maximum 8 payload bits. Receiver must terminate, correct, or propagate the error. Generate deterministic odd parity on the Master VIP for all signals. Negative scenarios: inject single-bit flips on TDATACHK and TLASTCHK independently to verify DUT Slave flags the error or drops the packet.

3. Test Cases & Mapping

Index	**Scenario**	**TR Mapping**
TC_MST_001	Basic single-cycle handshake transfer with full TKEEP and no interleaving.	REQ_MST_01, REQ_MST_02
TC_MST_002	Back-to-back zero-delay packet bursting.	REQ_MST_01, REQ_MST_02, REQ_MST_05
TC_MST_003	Force TVALID drop before TREADY assertion (Negative Test) to test VIP protocol monitor bounds.	REQ_MST_01
TC_MST_004	Heavy stream interleaving (4 unique TIDs) changing on a per-beat boundary.	REQ_MST_03
TC_MST_005	Sparse stream generation. Randomized TKEEP with mixed TSTRB mapping position bytes.	REQ_MST_04
TC_MST_006	Illegal Byte Qualifier (Negative Test). Generate TSTRB=1 while TKEEP=0 on random byte lanes.	REQ_MST_04
TC_MST_007	Null packet termination. Send TLAST=1 with all TKEEP=0.	REQ_MST_05
TC_MST_008	Asynchronous reset mid-transfer. Assert ARESETn while TVALID=1 and TREADY=0.	REQ_MST_06
TC_MST_009	Assert TWAKEUP exactly 1 cycle before TVALID for an entire multi-beat packet.	REQ_MST_07
TC_MST_010	Parity injection (Negative Test). Flip bit 0 of TDATACHK to create even parity.	REQ_MST_08
TC_MST_011	Parity corruption on control signals. Flip TLASTCHK during packet boundary.	REQ_MST_08

TC_MST_012	Maximum latency TREADY acceptance. DUT inserts 50 wait cycles dynamically.	REQ_MST_02

4. Functional Coverage Groups

Group Name	Description
cg_axi_handshake	Coverage over TVALID/TREADY timing dependencies and delays.
cg_axi_packet_boundaries	Cross coverage of TLAST with TDATA sizes and TKEEP sparsity.
cg_axi_interleaving	Number of concurrent TIDs active and interleaved beat frequencies.
cg_axi5_wakeup	Timing delta between TWAKEUP assertion and TVALID.
cg_axi5_parity	Tracking error injections across all supported Check_Type signals.

5. Interface Signals

Signal Name	Direction	Width	Description
ACLK	Input	1	Global clock signal.
ARESETn	Input	1	Global active-LOW reset.
TVALID	Output	1	Master indicates valid payload.
TREADY	Input	1	Slave indicates readiness to accept.
TDATA	Output	DATA_WIDTH	Primary payload. Parametrized (multiple of 8).
TSTRB	Output	DATA_WIDTH/8	Data byte/Position byte qualifier.
TKEEP	Output	DATA_WIDTH/8	Payload validity byte qualifier (null byte removal).
TLAST	Output	1	Packet boundary demarcation.
TID	Output	TID_WIDTH	Stream identifier for interleaving.
TDEST	Output	TDEST_WIDTH	Target destination routing descriptor.
TUSER	Output	TUSER_WIDTH	Sideband user-defined signaling.
TWAKEUP	Output	1	(AXI5) Wake-up indicator crossing clock domains.
TVALIDCHK	Output	1	(AXI5) Odd parity check for TVALID.
TREADYCHK	Input	1	(AXI5) Odd parity check for TREADY.
TDATACHK	Output	DATA_WIDTH/8	(AXI5) Odd parity mapped 1-bit per byte of TDATA.
TSTRBCHK	Output	ceil(DATA_WIDTH/64)	(AXI5) Odd parity for TSTRB.

TKEEPCHK	Output	ceil(DATA_WIDTH/64)	(AXI5) Odd parity for TKEEP.
TLASTCHK	Output	1	(AXI5) Odd parity for TLAST.
TWAKEUPCHK	Output	1	(AXI5) Odd parity for TWAKEUP.

6. Verification Environment Structure

axi_stream_master_vip_tb/

```
|-- master_agent/
|    |-- axi_stream_master_driver.sv
|    |-- axi_stream_master_monitor.sv
|    |-- axi_stream_master_sequencer.sv
|    |-- axi_stream_master_agent_config.sv
|    |-- axi_stream_master_callback.sv
|    |-- axi_stream_master_agent.sv
|-- env/
|    |-- axi_stream_master_scoreboard.sv
|    |-- axi_stream_master_env_config.sv
|    |-- axi_stream_master_env.sv
|-- sequences/
|    |-- axi_stream_master_base_sequence.sv
|    |-- axi_stream_master_seq_item.sv
|-- top/
     |-- axi_stream_master_tb_top.sv
     |-- axi_stream_master_test.sv
     |-- axi_stream_master_defines.sv
```

7. SystemVerilog Implementation

axi_stream_master_seq_item.sv (Architecture Blueprint)

```
COMPONENT: axi_stream_master_seq_item
EXTENDS: uvm_sequence_item
MEMBERS:
  - rand bit [DATA_WIDTH-1:0] TDATA
  - rand bit [(DATA_WIDTH/8)-1:0] TSTRB
  - rand bit [(DATA_WIDTH/8)-1:0] TKEEP
  - rand bit TLAST
  - rand bit [TID_WIDTH-1:0] TID
  - rand bit [TDEST_WIDTH-1:0] TDEST
  - rand bit [TUSER_WIDTH-1:0] TUSER
  - rand bit TWAKEUP
  - rand int unsigned delay_cycles
METHODS & LOGIC:
  - constraints: [delay_cycles within 0..15, TSTRB must be 0 where TKEEP is 0]
```

axi_stream_master_driver.sv (Architecture Blueprint)

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```
COMPONENT: axi_stream_master_driver
EXTENDS: uvm_driver #(axi_stream_master_seq_item)
CONFIG_REF: axi_stream_master_agent_config
MEMBERS:
  - virtual axi_stream_vif vif
METHODS & LOGIC:
  - build_phase: [Retrieve agent config and assign virtual interface]
  - run_phase: [
      Wait for reset release;
      Loop forever: get_next_item, drive TWAKEUP if configured,
      wait delay_cycles,
      drive TDATA, TKEEP, TSTRB, TLAST, TID, TDEST, TUSER;
      calculate and drive odd parity explicitly;
      assert TVALID;
      wait until TREADY is sampled HIGH;
      drop TVALID or process next item;
      item_done();
    ]
```